

UBL SageMath VerificationReproducibility Supplement

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Category: Computational Verification Report

1. Overview

This document provides a complete record of the SageMath computations used to verify the numerical results in the Universal Baseline Level (UBL) paper. All results are derived from standard physical constants with no external data dependencies. The script may be run in any SageMath environment (version 9.0 or later) to reproduce the output exactly.

2. Software and Environment

- Language:** SageMath (Python-compatible)
- Python standard library module used:** `math`
- No external packages required**
- Tested output format:** floating-point (machine precision)

3. Physical Constants Used

Constant	Symbol	Value (with units)
Speed of light	c	299,792,458 m/s

Constant	Symbol	Value (with units)
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$ J·s
Gravitational constant	G	6.67430×10^{-11} m ³ kg ⁻¹ s ⁻²
Cosmological constant	Λ	1.11×10^{-52} m ⁻²

4. Computed Quantities

Quantity	Description
Planck energy density (ρ_{Planck})	$c^5 / (\hbar G^2)$
Vacuum energy density (ρ_{vac})	$\Lambda c^2 / (8\pi G)$
Universal Baseline Level (UBL)	$\rho_{\text{vac}} / \rho_{\text{Planck}}$
$\log_{10}(\text{UBL})$	Base-10 logarithm of UBL
UBL scale table	UBL and $\log_{10}(\text{UBL})$ for seven reference energy densities
Effective Higgs cutoff (Λ_{eff})	$\text{UBL}^{1/4} \cdot \sqrt{(c^5 / (\hbar G))}$
EW/Gravity hierarchy ratio	$\text{UBL}_{\text{EW}} / \text{UBL}$ where $\text{UBL}_{\text{EW}} = 10^{-40}$
QCD step difference	$\log_{10}(\text{UBL}_{\text{QCD}}) - \log_{10}(\text{UBL})$ where $\text{UBL}_{\text{QCD}} = 10^{-65}$

5. SageMath Source Code

The following script reproduces all numerical results.

Python / SageMath

```
# Universal Baseline Level (UBL) Verification Script # This script reproduces
all numerical results in the UBL paper # using only standard physics
constants and definitions. # Load constants c = 299792458
# speed of light (m/s) hbar = 1.054571817e-34 # reduced Planck
constant (J·s) G = 6.67430e-11 # gravitational constant (m^3
kg^-1 s^-2) Lambda = 1.11e-52 # cosmological constant (m^-2)
import math # 1. Planck energy density rho_planck = c^5 / (hbar * G^2) # 2.
Vacuum energy density from cosmological constant rho_vac = Lambda * c^2 / (8
* math.pi * G) # 3. Universal Baseline Level (UBL) UBL = rho_vac /
```

```

rho_planck # 4. log10(UBL) log10_UBL = math.log10(UBL) # 5. UBL scale table
verification_densities = { "Planck": 5.16e96, "GUT": 1e88,
"Electroweak": 1e56, "QCD": 1e31, "Atomic": 1e16, "Biological":
1e-3, "Cosmic Vacuum": 6e-27 } UBL_table = {} for name, rho in
densities.items(): u = rho / rho_planck UBL_table[name] = (u,
math.log10(u)) # 6. Effective Higgs cutoff Lambda_eff = UBL^(1/4) *
math.sqrt(c^5 / (hbar * G)) # 7. Electroweak vs gravitational hierarchy
UBL_EW = 1e-40 ratio_hierarchy = UBL_EW / UBL # 8. QCD scale step difference
UBL_QCD = 1e-65 step_QCD = math.log10(UBL_QCD) - math.log10(UBL) # Print
results print("Planck energy density:", rho_planck) print("Vacuum energy
density:", rho_vac) print("UBL:", UBL) print("log10(UBL):", log10_UBL)
print("\nUBL Scale Table:") for k,v in UBL_table.items(): print(k, "=>
UBL:", v[0], " log10:", v[1]) print("\nEffective Higgs cutoff (Lambda_eff):",
Lambda_eff) print("EW/Gravity hierarchy ratio:", ratio_hierarchy) print("QCD
step difference:", step_QCD)

```

6. Verified Output

Running the script above in SageMath produces the following output exactly.

Console Output

```

Planck energy density: 5.15484850640341e96 Vacuum energy density:
5.94728569105370e-27 UBL: 1.15372657094887e-123 log10(UBL): -
122.93789710521712 UBL Scale Table: Planck => UBL: 1.00099934917393 log10:
0.00043379511142230905 GUT => UBL: 1.93992121932932e-9 log10: -
8.71221590651579 Electroweak => UBL: 1.93992121932932e-41 log10: -
40.71221590651579 QCD => UBL: 1.93992121932932e-66 log10: -65.7122159065158
Atomic => UBL: 1.93992121932932e-81 log10: -80.7122159065158 Biological =>
UBL: 1.93992121932932e-100 log10: -99.7122159065158 Cosmic Vacuum => UBL:
1.16395273159759e-123 log10: -122.93406465613215 Effective Higgs cutoff
(Lambda_eff): 3.41850751361282e12 EW/Gravity hierarchy ratio:
8.66756496019296e82 QCD step difference: 57.93789710521712

```

7. Notes on Reproducibility

- The caret operator (^) is used for exponentiation per SageMath syntax; substitute ** if running in plain Python.
- Floating-point results may vary at the last digit depending on platform arithmetic.
- All constants are hardcoded; no network access or external data is required.